TT-P-001385 (GSA-FSS)
September 17, 1976
SUPERSEDING
(see 6.3)

INTERIM FEDERAL SPECIFICATION

PRIMER COATING, SYNTHETIC, GRAY (FOR METAL EQUIPMENT)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two classes of fast drying primer suitable primarily for application to metal equipment.

1.2 Classification. The primer shall be of the following classes as specified in the invitation for bids, contract or purchase order.

Class 1 - Conventional ready-mixed form, in bulk containers, suitable primarily for spray application.
Class 2 - Primer in pressurized dispensers for touch-up of areas to which the class 1 primer has been applied.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

SS-L-30 - Lath, and Board Products, Gypsum.
TT-E-001384 - Enamel, Gloss, Synthetic, for Metal Equipment.
TT-T-001386 - Thinner, Synthetic Enamel (for Low and High Temperature Spray Application).
UU-C-282 - Chipboard.
LLL-B-1168 - Building Board, Hard Pressed, Vegetable Fiber (Laminated).
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials, Packaging, Packing, and Marking of.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials, Methods of Inspection, Sampling, and Testing.
Fed. Std. No. 595 - Colors.

FSC 8010

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards.

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Chemical Society (ACS) Standards:

Reagent Chemicals (Fifth Edition, 1974).

(Application for copies should be addressed to the American Chemical Society, 1155 Sixteenth Street, N.W., Washington, DC 20036.)

American Society for Testing and Materials (ASTM) Standards.

D 86 - Distillation of Petroleum Products, Test for.
D 154 - Varnishes, Testing.
D 523 - Specular Gloss, Test for.
D 562 - Consistency of Paints Using the Stormer Viscosimeter, Test for.
D 609 - Preparation of Steel Panels for Testing Paint, Varnish, Lacquer, and Related Products.
D 822 - Light- and Water-Exposure Apparatus (Carbon-Arc Type) for Testing Paint, Varnish, Lacquer, and Related Products on Test Panels.
D 1210 - Fineness of Dispersion of Pigment-Vehicle Systems, Test for.
D 1640 - Drying, Curing, or Film Formation of Organic Coatings at Room Temperature, Tests for.
D 2244 - Instrumental Evaluation of Color Difference of Opaque Materials.
D 3335 - Low Concentrations of Lead in Paint by Atomic Absorption Spectroscopy, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

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Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials. The primer shall be processed with suitable hiding and extender pigments, a vehicle consisting of synthetic resin, suitable volatile solvents, and such additives as drying aids, wetting agents, antisettling agents, antiflooding agents, plasticizers, stabilizers, and propellant as may be required. Polychlorinated biphenyls shall not be used in these products.

3.1.1 Volatile solvent (class 1 only). The volatile solvent, when tested as specified in 4.4.1, shall conform by volume to the requirements controlling the emission of solvents into the atmosphere as called out in (a), (b), (c), (d), and (e).

(a) Total of aldehydes and branched chain ketones: 20 percent maximum.
(b) Total of aromatic compounds with eight or more carbon atoms to the molecule except ethylbenzene. 8 percent maximum.
(c) Total of ethylbenzene and toluene. 20 percent maximum.
(d) Total of solvents with olefinic or cyclo-olefinic unsaturation. 5 percent maximum.
(e) Total of (a) + (b) + (c) + (d) = 20 percent maximum.

3.1.2 Infrared spectrophotometric analysis. When the nonvolatile vehicle is tested as specified in 4.4.7, band positions, shapes, and relative intensities in each spectrum shall match the spectrum in figure 1.

3.1.3 Propellant (class 2). The propellant shall be a hydrocarbon, a blend of hydrocarbons, halogenated hydrocarbon(s), or blend of hydrocarbon(s) and halogenated hydrocarbon(s), as required to ensure that the pressurized spray primer conforms to the requirements specified herein. Vinyl chloride shall not be used as a propellant.

3.2 Quantitative requirements. When tested as specified in 4.4, the quantitative requirements shall be as specified in table I.

TABLE I. Quantitative requirements

Characteristics	Requirements			
	Class 1		Class 2	
	Min.	Max.	Min.	Max.
Nonvolatile content (percent by weight of primer)	54.0	----	-----	----
Nonvolatile vehicle (percent by weight of vehicle)	35.0	----	-----	----
Pigment (percent by weight of primer)	28.0	----	-----	----
Fineness of grind	6	----	6	----
Consistency (KU: shearing rate 200 r.p.m.)	60	72	-----	----
Weight per gallon (pounds)	10.0	10.5	-----	----
Drying time:				
Set to touch (minutes)	-----	10	-----	10
Dry hard (hours)	-----	6	-----	6
Dry for recoat (hours)	-----	8	-----	8
Dry for taping (hours)	-----	3	-----	3
Free from after tack (hours)	-----	24	-----	24
Dry opacity (contrast ratio):				
4.7 ml per square foot	0.98	----	0.98	----
Accelerated weathering for 300 hours:				
60° specular gloss				
Before weathering	-----	5	-----	5
After weathering	-----	6	-----	6
Color difference (ΔE)				
After weathering	-----	2.0	-----	2.0
Chalking				
After weathering	none	none	none	none

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TABLE I. Quantitative requirements (con.)

Characteristics	Requirements			
	Class 1		Class 2	
	Min.	Max.	Min.	Max.
Hardness (Sward) After drying 1 week	10	----	10	----
Net contents, by weight, of dispenser (grams)	----	----	368	----
Total nonvolatile content of primer in dispenser (grams)	----	----	59.0	----
Pigment (percent by weight of the total nonvolatile contents)	----	----	37.0	----
Nonvolatile vehicle (percent by weight of total nonvolatile contents)	----	----	62.0	----
Lead (percent by weight of total nonvolatile)	----	0.5	----	0.5

3.3 Pressurized dispenser (class 2).

3.3.1 Container. The container shall be a nominal 16-ounce commercial type metal pressure container generally known as an aerosol container of such construction as to assure acceptance of the finished package by common carriers operating in interstate commerce. The capacity of the container shall be sufficient to contain the specified net contents of the product with outage in accordance with generally recognized safe commercial practice. Each container shall be labeled as stated in section 5. The contents shall not react with nor be reacted upon by the interior surface of the dispenser or any of the dispenser components.

3.3.2 Dispensing valve. The valve shall have a spray head which can be removed without releasing pressure from the aerosol container. The valve shall have a metering orifice accessible to cleaning with a common pin after removal of the spray head. The removable spray heads shall contain orifices of such dimensions as to produce spraying properties as required in 3.3.5. The lower end of the dip tube shall not elongate and touch the bottom of the can and shall have a maximum clearance of 3 mm (1/8 inch) from the bottom rim of the can after storage for 1 week in the product (which allows for swelling of the tube).

3.3.3 Agitator. Each dispenser shall contain one spherical agitator either 3/8, 5/8, or 3/4 inch in diameter, or the equivalent volume in smaller spheres, such one or more glass marbles or one or more steel balls, or one or more irregularly shaped steel pieces, such as rivets.

3.3.4 Cover cap. The valve shall be protected from accidental functioning and damage by a press-fitting metal or plastic cover cap which shall be removable and replaceable. The color of the cover cap shall be an approximate match to the color of the primer coating in the container.

3.3.5 Dispenser package.

3.3.5.1 Internal pressure. When tested as specified in 4.4.19, the pressure shall be not more than 110 pounds per square inch, gauge (p.s.i.g.) at 54°C (130°F).

3.3.5.2 Spray pattern. When tested as specified in 4.4.18.2.1, the dispenser shall deliver a spray pattern approximately 3 inches wide when held at a distance of 5 to 10 inches from a vertical surface and spraying is done at 20 ± 1°C (73 ± 2°F). The spray pattern so deposited shall be uniform on the smooth vertical surface specified. The primer must retain the spray pattern requirements through 98 percent exhaustion of the container (see 3.3.5.3).

3.3.5.3 Spray delivery. When tested as specified in 4.4.18.2.2, the contents of the dispenser container shall be sprayable when stabilized at temperatures from 16 to 32°C (60 to 90°F). A minimum of 98 percent by weight of content shall be sprayable in a pattern such as to deposit a uniform coating on a smooth vertical surface. The film shall dry free of orange peel, sags, specks, seeds, or any other surface irregularities. Leftover primer of more than 2 percent shall constitute a defect.

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3.4 Qualitative requirements.

3.4.1 Condition in container (class 1 only). A freshly opened full container of the primer, when tested as in 4.4.15, shall be free from grit, seeds, skins, lumps, abnormal thickening, or livering, and shall show no more pigment settling or caking than can be readily reincorporated to a smooth homogeneous state.

3.4.2 Storage stability (class 1 only). A 3-quarter filled, closed 8-ounce glass jar of the primer shall show no skinning when tested as specified in 4.4.16.

3.4.3 Compatibility with thinner (class 1 only). When tested as specified in 4.4.17, the primer shall show no evidence of coagulation, separation, precipitation. or other symptoms of incompatibility with thinner.

3.4.4 Spraying properties (class 1 only). When tested as specified in 4.4.18.1, the primer shall spray satisfactorily in all respects, without running, sagging, or streaking. The dried film shall show no dusting, mottling, or color separation, and shall have a smooth, uniform finish free from seeds.

3.4.5 Color. Unless otherwise specified, the color of the primer shall match number 36306 of Fed. Std. No. 595, when tested as specified in 4.4.10.

3.4.6 Flexibility. When tested as specified in 4.4.12, a dry film of the primer shall withstand bending without cracking and flaking.

3.4.7 Resistance to water immersion. When tested as specified in 4.4.13, a dry film of the primer shall resist degradation when immersed in distilled water.

3.4.8 Lifting (class 2 only). When tested as specified in 4.4.18.2.3, the primer coating shall not lift a dry film of itself or a dry film of enamel conforming to TT-E-001384. When tested in a similar manner, a dry film of the primer coating shall not be lifted by enamel conforming to TT-E-001384.

3.4.9 Shelf storage life.

3.4.9.1 Class 1. When stored as specified in 4.4.20, the primer shall meet the requirements in 3.4.2.2 and 3.4.5, and the consistency and drying times shall not vary more than plus or minus 10 percent of the values in table I.

3.4.9.2 Class 2. When stored as specified in 4.4.20, the primer shall conform to the following requirements:

(a) Valve condition (see 3.3.2).
(b) Pressure (see 3.3.5.1).
(c) Spraying properties (see 3.3.5.2 and 3.3.5.3).
(d) Drying times (see table I).
(e) Contrast ratio (see table I).
(f) Color (see 3.4.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection of preparation for delivery.

4.2.1 Class 1. The packaging packing, and marking shall be inspected in accordance with PPP-P-1892.

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4.2.2 Class 2. The packaging, packing, and marking shall be inspected for conformance with the requirements of 3.3, 5.2.2, and 5.3 as applicable. Inspection lots and sampling shall be in accordance with MIL-STD-105, inspection level I. Defects indicating non-conformance with 3.3, 5.2.2, and 5.3 shall be inspected using Acceptable Quality Level (AQL) of 1.5 percent defective. Defects indicating inadequate warning labeling (see 5.3.3 (g), (h), (i), and (j)) shall be inspected using an AQL of 0.015 percent defective.

4.3 Inspection of the end item.

4.3.1 Class 1. For end item inspection not involving tests, the filled unit containers of the primer shall be assembled into lots as specified in MIL-STD-105. In MIL-STD-105 the definition, "essentially the same condition", shall be interpreted to mean all the filled unit containers from a manufacturer's batch, which is defined as the end product of all raw materials mixed, blended, or processed in a single operation. Inspection shall be made in accordance with MIL-STD-105, inspection level S-2, with an AQL of 4.0 percent defective.

4.3.2 Class 2. For end item inspection not involving tests, the filled pressurized dispensers shall be assembled into lots as specified in MIL-STD-105. In MIL-STD-105, the lot definition, "essentially the same condition", shall be interpreted to mean all the filled pressurized dispensers offered for delivery at one time. Inspection shall be in accordance with MIL-STD-105, level S-2, with an AQL of 4.0 percent defective.

4.3.3 Sampling of the end item for tests.

4.3.3.1 Class 1. For purposes of sampling, the lot (see 4.3.1) shall be expressed in units of gallons. Samples from lots shall be taken in accordance with MIL-STD-105, using inspection level S-2 and an AQL of 2.5 percent defective.

4.3.3.2 Class 2. For purposes of sampling, the lot (see 4.3.2) shall be expressed in units of filled pressurized dispensers. Samples from lots shall be taken in accordance with MIL-STD-105, using inspection level I and an AQL of 1.5 percent defective.

4.3.4 The supplier shall submit to the contractor a certificate of compliance indicating that the primer complies with the shelf storage life requirement as specified in 3.4.9. When certificates of compliance are submitted, the Government reserves the right to test such items to determine the validity of the certificate.

4.4 Test procedures. The primer shall be tested in accordance with table II, and as hereinafter specified, to determine compliance with the applicable requirements in section 3. All test reports shall contain the individual values utilized in expressing the final result. Failure of any test, or noncompliance with any requirement shall be cause for rejection of the sample and, therefore, the lot. Unless otherwise specified, all tests shall be conducted at standard conditions, 23 $\pm$ 1°C (73 $\pm$ 2°F) and 50 $\pm$ 5 percent relative humidity.

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TABLE II. Index of tests

Test	Method	Test ref.	Rqmt. ref.
Nonvolatile content	4044 1/	4.4.4	Table I
Nonvolatile vehicle	4032 1/	4.4.5	Table I
Pigment	4022 1/	4.4.6	Table I
Fineness of grind	D 1210 2/	---	Table I
Consistency	D 562 2/	---	Table I
Weight per gallon	4184 1/	---	Table I
Drying time	D 1640 2/	4.4.8	Table I
Dry for taping	---	4.4.8.3	Table I
Contrast ratio	4121 1/	4.4.9	Table I
Specular gloss	D 523 2/	4.4.11	Table I
Accelerated weathering	D 822 2/	4.4.11	Table I
Color	4250 1/	4.4.10	3.4.5
Color difference	D 2244 2/	4.4.11	Table I
Chalking	D 822 2/	4.4.11	Table I
Hardness	---	4.4.14	Table I
Preparation of steel panels	D 609 2/	4.4.18.2.2	3.3.6.4
Preparation of tin panels	2012 1/	4.4.11	Table I
		4.4.12	Table I
		4.4.13	Table I
Flexibility	6221 1/	4.4.12	3.4.6
Spraying properties	4331 1/	4.4.18.1	3.4.4
Lifting	6251 1/	4.4.18.2.3	3.4.8
Pressure	---	4.4.19	3.3.6.1
Spray pattern	---	4.4.18.2.1	3.3.6.2
Spray delivery	---	4.4.18.2.2	3.3.6.3
Condition in container	3011 1/	4.4.15	3.4.1
Storage stability	D 154 2/	4.4.16	3.4.2
Compatibility with thinner	---	4.4.17	3.4.3
Immersion resistance	---	4.4.13	3.4.7
Shelf storage life	---	4.4.20	3.4.9
Lead content	D 3335 2/	---	Table I
Infrared spectrophotometry	---	4.4.7	3.1.2
Solvent distillation	D 86 2/	4.4.1	3.1.1
Solvent analysis	7356 1/	4.4.1.1	3.1.1
		4.4.1.2	3.1.1
Net contents	---	4.4.3	Table I

1/ Federal Test Method Std. No. 141.
2/ American Society for Testing and Materials (ASTM).

4.4.1 Solvent analysis (class 1). The solvent shall be distilled from 100 ml of the primer according to ASTM D 86.

4.4.1.1 Solvent composition. The solvent composition shall be determined in accordance with method 7356 of Fed. Test Method Std. No. 141 and 4.4.1.2 to determine compliance with the requirement of 3.1.1.

4.4.1.2 Aromatic and oxygenated solvents. The 1.83 m column shall be installed and the operating conditions described in method 7356 shall be followed. About 3 microliters of the isolated distillate shall be injected and the chromatogram scanned. The aliphatic solvents will emerge within 1 minute and the complete chromatogram should develop in about 5 minutes. From the position of the peaks observed on the chromatogram, select an internal standard such as cyclopentanol or cyclohexanol that will not interfere with other peaks. Six-tenths of a milliliter of the internal standard shall be added to 3 ml of the distillate. The sample shall be analyzed according to the above procedure. Peaks emerging after

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1 minute are aromatic solvents and any oxygenated solvents that may be present. The percent of aromatic and oxygenated solvents shall be calculated as follows:

$$\text{Percent aromatic and oxygenated solvents, v/v} = \frac{A \times B}{C \times D}$$

Where: A = area of aromatic and oxygenated solvents.
B = percent of internal standard added; in this case, 20.
C = area of internal standard.
D = The calibration factor for the internal standard. This factor is dependent on the internal standard used and on the performance of the chromatograph, and should be determined daily.

4.4.2 Test concentrate (class 2). Where specified, tests on the class 2 primer shall be conducted on a test concentrate prepared as follows: With the container well agitated and in an upright position, pierce the top with a very fine hole (with a needle point), which will allow the propellant to escape (conditioning at low temperature for 1 hour will facilitate this). After a few minutes, the pierced hole can be enlarged, and as soon as there is no further evidence of gas escaping, the top of the container can be cut open. Allow to stand until the contents are at room temperature, then transfer the contents quantitatively into a tared beaker using several washes of acetone. Evaporate on a water bath until no more propellant is liberated and until the contents are less than 40 percent of the original weight of the contents including propellant. Then readjust to exactly 40 percent with n-butyl acetate.

4.4.2.1 For the qualitative tests, the manufacturer may use the primer concentrate as it exists before packaging and pressurization, provided it is adjusted as specified in 4.4.2.

4.4.3 Total net contents (class 2). The total net contents shall be determined by calculating the difference in weight between the filled container and the completely empty one (see 4.4.2 for procedure of emptying). Wash out interior of can with acetone and dry. (Note that the weight of the agitators must be included in the tare.) Using the completely emptied container as tare, additional checks can be made on filled containers. At least two containers shall be checked from each lot sampled since variations in weight of empty cans can be found. A check on full and empty cans shall be made on the same cans.

4.4.4 Nonvolatile content.

4.4.4.1 Class 1. Test for nonvolatile content of the primer, as received, in accordance with method 4044 of Federal Test Method Std. No. 141.

4.4.4.2 Class 2. Use method 4044, Federal Test Method Std. No. 141, on the test concentrate. Run test in duplicate. The nonvolatile figure obtained is multiplied by 0.40 to obtain the total solids in the pressurized container as received.

4.4.5 Nonvolatile vehicle.

4.4.5.1 Class 1. Test for nonvolatile vehicle in accordance with method 4032 of Federal Test Method Std. No. 141 using the following mixture for extraction:

(a) 1 volume petroleum ether (ACS Reagent).
(b) 1 volume toluene (ACS Reagent).
(c) 1 volume acetone (ACS Reagent).
(d) 3 volumes absolute ethyl alcohol (ACS Reagent).

4.4.5.2 Class 2. Test for nonvolatile vehicle from the test concentrate described in 4.4.2, by method 4032 of Federal Test Method Std. No. 141, using the following mixture for extraction:

(a) 1 volume petroleum ether (ACS Reagent).
(b) 1 volume toluene (ACS Reagent).
(c) 1 volume acetone (ACS Reagent).
(d) 9 volumes hot absolute ethyl alcohol (ACS Reagent).

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4.4.6 Pigment.

4.4.6.1 Class 1. Test for pigment by method 4022 of Federal Test Method Std. No. 141, using the extraction mixture specified in 4.4.5.1. Calculate pigment on the basis of the total weight of the primer.

4.4.6.2 Class 2. Test for pigment on the test concentrate by method 4022 of Federal Test Method Std. No. 141, using the extraction mixture described in 4.4.5.2. Calculate pigment on the total nonvolatile content obtained as in 4.4.4.2.

4.4.7 Infrared spectrophotometric analysis. Test a film of the nonvolatile vehicle, obtained by separation of the primer as specified in 4.4.5, and oven dried for 1 hour at 105°C, in an infrared spectrophotometer.

4.4.8 Drying time.

4.4.8.1 Class 1. Except for the dry-for-taping time, test for all drying times by method ASTM D 1640, applying the primer to a wet film thickness of 0.038 mm (0.0015 inch).

4.4.8.2 Class 2. Prepare four panels for drying time by spraying from the dispenser on each panel a dry film of approximately 0.012 mm (0.0005 inch). Test two of the panels for drying times specified in table I in accordance with method ASTM D 1640. Allow the other two panels to dry for 24 hours, then spray on each a second coat of approximately 0.012 mm (0.0005 inch) of the primer coating. Test the second coat for drying times specified in table I, in accordance with ASTM D 1640.

4.4.8.3 Dry for taping. Apply the primer to a smooth glass panel in such manner as to give a dry film thickness of 0.051 ± 0.005 mm (0.002 ± 0.0002 inch). After the film has dried in a horizontal position for 3 hours, apply across the entire width of the film a piece of 25 mm (1 inch) wide masking tape. Roll the tape smoothly and uniformly with an 11 kg (5 lb) roller. Remove the tape with a smooth rapid pulling. Any removal or deformation of the primer film shall constitute failure to pass the dry-for-taping test.

4.4.9 Contrast ratio.

4.4.9.1 Class 1. Test in accordance with method 4121 of Federal Test Method Std. No. 141.

4.4.9.2 Class 2. Apply a wet film, 0.051 mm (0.002 inch) in thickness, of the test concentrate obtained in 4.4.2 to a black and white hiding power chart. Application, drying, and testing of the film shall be in accordance with method 4121 of Federal Test Method Std. No. 141.

4.4.10 Color. Apply the primer, at a wet film thickness of 0.076 mm (0.003 inch), to a clean, smooth, white panel. Allow the film to dry for 24 hours in a horizontal position and determine the color in accordance with method 4250 of Federal Test Method Std. No. 141.

4.4.11 Accelerated weathering. Using tin panels as specified in method 2012 of Federal Test Method Std. No. 141, prepare six panels by applying sufficient primer to a dry film thickness of 0.051 ± 0.005 mm (0.002 ± 0.0002 inch). Air dry the panels at 23 ± 1°C (74 ± 2°F) for 96 hours. Expose three of the panels to accelerated weathering for 300 hours in accordance with ASTM D 822. Air dry the panels for 2 hours at standard conditions (see 4.4) and examine for chalking. Wash the weathered panels with running water and a thoroughly degreased lamb's wool pad. Dry the panels with clean cheesecloth, then air dry for 2 hours as above. Test the weathered and unweathered panels for gloss in accordance with ASTM D 523. Test the weathered and unweathered panels for color difference in accordance with ASTM D 2244. Report values as the average for the triplicate panels.

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4.4.12 Flexibility (cold cracking) The flexibility tests shall be as determined by method 6223 of Federal Test Method Std. No. 141. Test panels shall be as specified in method 2012 of the standard. Draw down films, to a dry thickness of 0.025 mm (0.001 inch), of the class 1 primer, as received, or the test concentrate (see 4.4.2) of the class 2 primer, air dry for 48 hours at standard conditions (see 4.4). The film shall withstand the specified cold cracking test when bent over a 6 mm (1/4 inch) rod.

4.4.13 Immersion resistance. Using a film applicator with 0.152 mm (0.006 inch) gap clearance, apply films of the as received class 1 primer, or test concentrate (see 4.4.2) of class 2 primer, on tin panels as specified in method 2012 of Federal Test Method Std. No. 141. Air dry for 24 hours at standard conditions (see 4.4), then immerse for 16 hours in distilled water at 23 ± 1°C (74 ± 2°F). Immediately upon removal from the water, the films shall show no wrinkling or blistering. After the panels are then air dried for 24 hours, the films shall show no more than slight dulling or whitening.

4.4.14 Hardness. Apply films of the primer 0.076 mm (0.003 inch), in wet film thickness, to two smooth glass panels and air dry in a horizontal position for 1 week at standard conditions (see 4.4). Test the films for hardness with a Sward rocker in accordance with instructions of the instrument manufacturer.

4.4.15 Condition in container (class 1 only). Determine condition in container by method 3011 of Federal Test Method Std. No. 141.

4.4.16 Storage stability (class 1 only). Determine skinning in a part full container by ASTM D 154.

4.4.17 Compatibility with thinner (class 1 only). Pour 100 ml of the well mixed primer into a 250-ml high-form glass beaker. Add 100 ml of thinner conforming to type I of TT-T-001386. Stir the primer and thinner thoroughly but without making bubbles. Immediately flow out a thin film of the mixture on a clear glass plate. Examine the film for nonuniformities such as small gelled particles, which will indicate coagulation of the primer vehicle and constitute failure of the requirements in 3.4.3. Small gelled particles may be differentiated from possible pigment flocculation by probing with a sharp-pointed metal object. After flowing out the films, allow the mixture in the beaker to remain undisturbed for 2 hours, and then examine the settled portion for possible coagulation of the primer vehicle. Any settling or lumping which is not easily redispersed by light stirring with a glass rod shall constitute failure to meet 3.4.3.

4.4.18 Spraying properties.

4.4.18.1 Class 1. Reduce the primer if necessary to spraying consistency according to the manufacturer's instructions using a thinner conforming to type I of TT-T-001386. Using any suitable, conventional, and commercially available spray equipment, spray the primer of proper consistency on a prepared wallboard panel conforming to SS-L-30 or LLL-B-1188. While spraying, hold the spray gun perpendicular to the panel and move it in straight lines back and forth across the panel, maintaining a distance of 6 to 10 inches between spray head and panel. Place the sprayed panel in a vertical position and observe the spray pattern for uniformity. Allow the sprayed film to dry, and examine it in accordance with method 4331 of Federal Test Method Std. No. 141 for compliance with 3.4.4.

4.4.18.2 Class 2.

4.4.18.2.1 Spray pattern. Condition the dispenser for 1 hour at 23 ± 1°C (74 ± 2°F). Use No. 1 kraft, wrapping paper, untreated, 90 pounds basis weight (24 inches by 36 inches, 500 sheets). Attach a sheet of this paper, approximately 1 yard square, in a vertical position to a wall or other stationary surface. Holding the pressurized dispenser about 5 to 10 inches from the surface, spray the primer on the kraft paper for approximately 10 seconds without moving the dispenser. Observe the primer pattern thus sprayed on the paper surface. If for any reason the spraying is not done in a manner so that an even, continuous flow of primer is delivered, the dispenser should be agitated and the test repeated. Not less than three dispensers of primer from each inspection lot shall be tested for conformance with 3.3.5.2.

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4.4.18.2.2 Spray delivery. Set a 4- by 12-inch steel panel, (prepared in accordance with ASTM D 609), at an angle of approximately 15° from the vertical. Hold the dispenser in an upright parallel position to the panel at a distance of 10 to 12 inches and with the valve fully open, move the dispenser horizontally from one side of the panel to the other using a stroking motion. Spray in overlapping horizontal strokes from the top to the bottom of the panel. Apply two thin coats (each coat approximately 0.012 mm (0.0005 inch) dried film thickness) one after the other. The second coat shall be applied by spraying vertically in overlapping strokes from side to side. After application of the second thin coat, place the panel in a vertical position, to dry for 5 minutes, and examine the film for smoothness of flowout, freedom from specks, bubbles, seeds, craters, or any apparent abnormalities. Continue spraying the contents to completion on a flat surface and note the spraying properties. Weigh the can. Open the can and wash out the left-over primer, then allow to dry for at least 2 hours. Weigh the can; then calculate the left-over primer as follows:

$$\text{Percent of leftover primer} = \frac{B-C}{A-C} \times 100$$

Where: A = weight in grams of full can.
B = weight in grams of empty can after spraying.
C = weight in grams of empty can thoroughly washed.

Check the results for compliance with the requirements of 3.3.5.3.

4.4.18.2.3 Lifting (class 2 only). By spraying from the pressurized dispenser, apply to each of two metal panels a film of the class 2 primer to a dry film thickness of 0.012 ± 0.002 mm (0.0005 ± 0.0001 inch), dry in a horizontal position for 24 hours at standard conditions (see 4.4). To one panel, then apply by spraying a second coat of the class 2 primer. Examine for lifting in accordance with method 6251 of Federal Test Method Std. No. 141. To the second panel, spray to a dry thickness of 0.025 ± 0.005 mm (0.001 ± 0.0002 inch), a film of enamel conforming to class 2 of TT-E-001384, and examine for lifting as above. To a third panel apply a film, to a dry film of 0.001 ± 0.0002 inch), of enamel conforming to class 1 or 2 of TT-E-001384. Dry for 24 hours as above, then spray a film of class 2 primer over the enamel to a dry film thickness of (0.001 ± 0.0002 inch). Examine for lifting as above.

4.4.19 Internal pressure (class 2 only).

4.4.19.1 Equipment. U.S. Gauge Company gage number 10897, or equivalent (calibrated to read 0 to 100 p.s.i.g.), with valve fitting for insertion into pressurized aerosol dispensers with a Sprayon Danvern valve, or equal.

4.4.19.2 Procedure. Condition the pressurized dispensers to be tested at the temperature specified in 3.3.5.1. Turn the dispenser upside down with the spray tip seated correctly in the valve. Clean the valve of primer by depressing the spray tip until only gas is emitted from the tip. Keeping the dispenser in the inverted position, push the gauge into the valve and immediately read the gauge. Pull the gauge from the dispenser before turning the dispenser to the upright position. Not less than six dispensers shall be tested from an inspection lot. Report the maximum, minimum, and average pressures for conformance with 3.3.5.1.

4.4.20 Shelf storage life. Store a 500-ml (1-pint) sample of the class 1 primer in a tightly sealed glass jar, or three filled pressurized dispensers undistrubed for 1 year from date of manufacture at standard conditions (see 4.4). Examine and test for compliance with 3.4.9 (see 4.1.1).

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5. PREPARATION FOR DELIVERY

5.1 Class 1 (conventional ready-mixed).

5.1.1 Packaging, packing, and marking. The primer coating shall be packaged, packed, and marked in accordance with PPP-P-1892. The level of packaging shall be A, B, or C, and the level of packing shall be A, B, or C, as specified (see 6.2). The primer coating shall be furnished in 1-pint cans, 1-quart cans, 1-gallon cans, 5-gallon pails, and 55-gallon drums (see 6.2).

5.2 Class 2 (primer in pressurized dispensers).

5.2.1 Packaging. (See 3.3).

5.2.2 Packing.

5.2.2.1 Level A. Twelve dispensers of the same description shall be packed in a close-fitting box conforming to PPP-B-636, V3c; or PPP-B-640, class 2. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. The gross weight of the V3c box shall not exceed the weight limitations of PPP-B-636.

5.2.2.2 Level B. Twelve dispensers of the same description shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. Separators the full height of the box shall be required conforming to UU-C-282, 0.040 inch minimum thickness or PPP-F-320. The box shall be closed in accordance with the appendix to PPP-B-636.

5.2.2.3 Level C. Primer coating, packaged as specified in 5.2.1, shall be packed to insure carrier acceptance and safe delivery to destination. Containers shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 Civil agencies. Packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Special marking. Legible commercial marking shall be acceptable. Cans overcoated to obliterate original marking shall not be acceptable. Marking shall be lithographed on the unit container or printed on a label securely affixed to the container and shall include the following:

(a) Name of material.
(b) Volume in units of US gallons (class 1).
(c) Net weight in ounces (class 2).
(d) Color number of Fed. Std. No. 595.
(e) Date of manufacture, in month and year.
(f) Date of first reinspection (Insert date one year after date of manufacture.).
(g) "Do not puncture or incinerate." (Class 2 only).
(h) "Keep away from fire or flame." (Class 2 only).
(i) "Use with adequate ventilation."
(j) "Store below 49°C (120°F)." (Class 2 only).

5.3.3.1 Shelf life marking. Markings specified in subparagraphs (e) and (f) of 5.3.3 shall be applied to intermediate and shipping containers.

6. NOTES

6.1 Intended use. This primer is used as an undercoat in coating systems of relatively fast drying enamels applied to automotive equipment, postal letter boxes, and other metal surfaces subject to sustained exterior exposure. The conventional ready mixed class 1 is intended for use in the total coating systems. The class 1 primer may be brushed, but it is suited more for spray application. The class 2 primer is intended to be compatible in the same coating systems as class 1, but it is for use primarily for touch-up application in small areas.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Color required (see 3.4.5).
(d) Level of packaging and level of packing required (see 5.1 and 5.2).
(e) Special marking required (see 5.3.3).
(f) Size of container required (see 5.1 and 3.3).

6.3 Supersession data. This specification supersedes General Services Administration Purchase Descriptions as follows:

TT-P-001385	GSA Purchase Description
Class 1	PD-FSS-5A
Class 2	PD-FSS-12.

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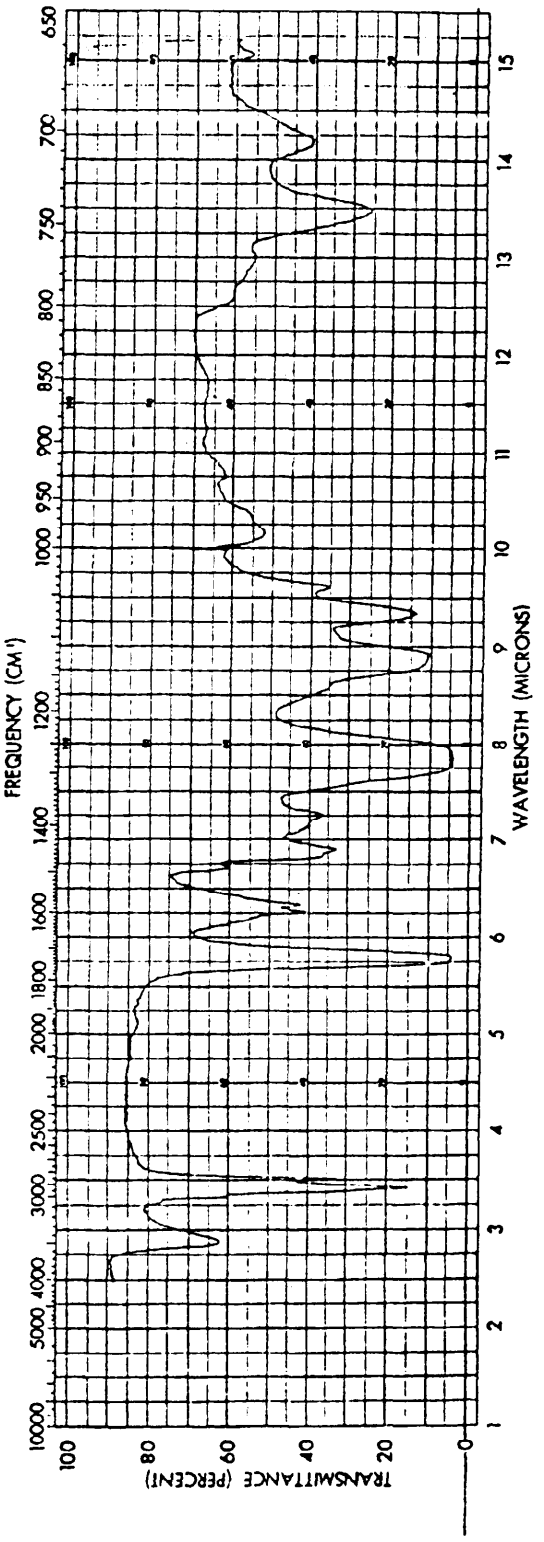


Figure 1 - Infrared Analysis of Nonvolatile Vehicle

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